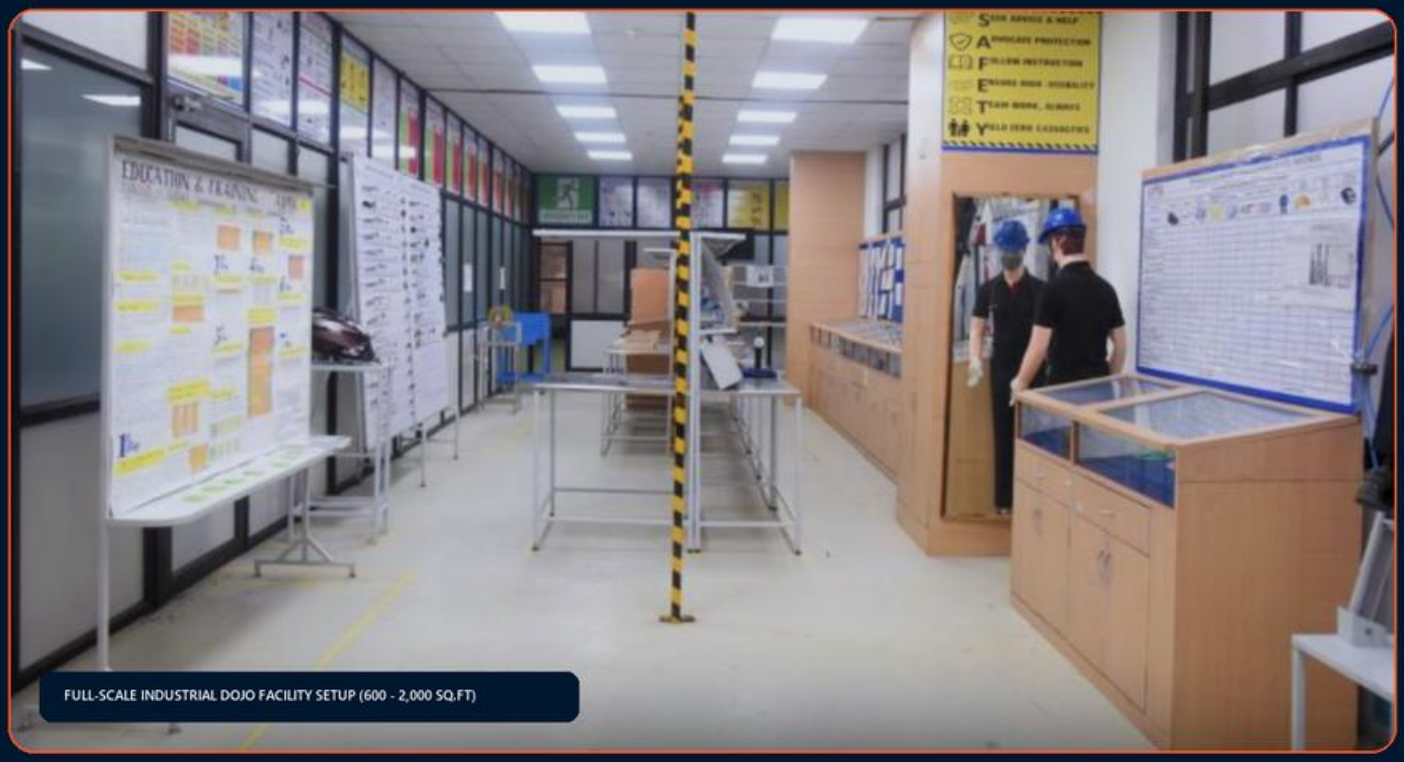
TETRAHEDRON
MANUFACTURING SOLUTIONS & INDUSTRIAL EXCELLENCE

DOJO TRAINING CENTER

Complete Turnkey Setup & Industrial Skill Academy

Engineering zero-defect manufacturing workforces: Turnkey spatial design, tactile simulation hardware, L1–L4 capability matrices, and Japanese lean shopfloor culture.



FULL-SCALE INDUSTRIAL DOJO FACILITY SETUP (600 - 2,000 SQ.FT)

80%
Faster Onboarding
5–7 days to takt speed

Zero
Training Scrap
100% simulated trial runs

60–80%
Near-Miss Drop
Experiential danger dome

3–6 Mo.
Capital Payback
Measured plant ROI payback

WHAT THIS TECHNICAL BROCHURE COVERS

01 **The Japanese Dojo Concept & Six Core Strengths**
Tactile repetition vs classroom forgetfulness, muscle memory, and safety culture.

02 **The Proven 6-Zone Blueprint & Spatial Planning**
Induction, Metrology, Safety Dome, Standard Work, TPM, and RCA Assessment.

03 **L1 to L4 Skill Matrix & Simulation Hardware Catalog**
Structured progression from novice to master trainer with CE-certified simulation pods.

04 **8-Week Turnkey Implementation Roadmap & Plant ROI**
From defect scrap Pareto audit and 3D CAD design to Train-the-Trainer accreditation.

05 **Enterprise Track Record & Client Partnerships**
100+ Dojos commissioned across India for Hero, Royal Enfield, Autoliv, Tata Motors, etc.

What is an Industrial Dojo Training Center?

THE JAPANESE PHILOSOPHY: BRIDGING THEORY & SHOPFLOOR EXECUTION

Originating from the Japanese martial arts concept of Dōjō ('Place of the Way'), an industrial Dojo is a dedicated, off-line capability center built directly within a manufacturing facility. It replaces passive classroom PowerPoint sessions with tactile, hands-on muscle-memory drills on realistic simulation rigs before operators are ever deployed to live production lines.

At Tetrahedron, we transform this concept into a turnkey engineering solution: designing the CAD spatial layout, fabricating custom simulation benches with CE safety interlocks, authoring bilingual Standard Work SOS/SOP sheets, and training certified internal Dojo Masters.

TRADITIONAL SHOPFLOOR ONBOARDING

- X 70% of classroom theory forgotten within 24 hours
- X High component scrap & tooling damage on live line
- X Assembly errors escape to customer (PPM spikes)
- X 30 to 45 days required for operator takt-time readiness
- X High safety risk & near-miss rate for fresh operators

TETRAHEDRON ZERO-DEFECT DOJO MODEL

- [OK] 100% muscle-memory retention through tactile repetition
- [OK] Zero live production scrap (practiced on reusable test jigs)
- [OK] Zero-defect habits instilled before touching live product
- [OK] Operators reach 100% takt speed in just 5 to 7 days
- [OK] 60%–80% reduction in plant near-misses via Danger Dome

THE 6 CORE STRENGTHS ENGINEERED BY TETRAHEDRON

Every simulation station is purpose-built to develop these foundational industrial capabilities:

01 Accelerated Skill Enhancement

Provides a controlled, focused environment for rapid learning and mastery of practical assembly tasks through structured repetition and immediate feedback.

KEY IMPACT: 80% faster cycle-time readiness

02 Bridging Theory & Practice

Translates engineering drawings and classroom guidelines into tactile muscle memory directly aligned with machine constraints and line takt times.

KEY IMPACT: Zero classroom disconnect

03 Building Lean Culture (TPS / 5S)

Instills Japanese principles of 5S, Kaizen, standardized work, and relentless waste elimination into operator daily habits from day one.

KEY IMPACT: 100% lean mindset adoption

04 Reinforcing Standardization

Ensures uniform understanding and strict adherence to Standard Operating Procedures (SOPs), SOS sheets, and ergonomic rhythms across all shifts.

KEY IMPACT: Zero operator variance

05 Effective Problem Solving

Develops critical thinking by equipping technicians with structured root-cause analysis tools: 5-Why, Fishbone (Ishikawa), and A3 problem-solving.

KEY IMPACT: Root cause, not symptom fixes

06 Improved Safety Awareness

Allows zero-risk experiential practice of dangerous pinch points, LOTO lockouts, and emergency interlocks without endangering workers.

KEY IMPACT: 60–80% drop in near-misses

THE FIVE-SENSE EXPERIENTIAL CALIBRATION FRAMEWORK

Traditional plants rely solely on visual inspection. Tetrahedron Dojos calibrate all five human sensory reflexes:

TOUCH	VISION	HEARING	KINETIC	REFLEX
Blind sorting of M6 vs M8 threads, torque 'click' feel, sheet burr detection	Lux-calibrated scratch detection, color variance, alignment gauges	Pneumatic micro-leak hiss detection, bearing rattle, torque limiter click	Ergonomic lifting posture, balanced hand reach, fatigue-free stance	Instant emergency-stop punch response, pinch-point avoidance habits

The Proven 6-Zone Dojo Blueprint

MODULAR, SCALABLE ARCHITECTURE FOR ANY MANUFACTURING SPACE

Whether allocating 600 sq.ft for a focused safety/assembly mini-dojo or 2,000+ sq.ft for a corporate academy, Tetrahedron partitions the space into 6 synergistic zones that seamlessly take a trainee from corporate culture to certified production readiness.

Zone 01 Culture, Vision & Visual Managem



Gateway to Quality & Lean Heritage

The entry gateway where operators absorb customer defect impact and plant pride before touching live equipment.

KEY EQUIPMENT & RIGS:

- Visual Performance Board Kiosks
- Cutaway Finished Assemblies
- Customer Defect 'Hall of Shame'
- 5S Workplace Standards Display

IMPACT: **Builds zero-defect pride before shopfloor entry**

Zone 02 Quality Assurance & Sensory Cali



Metrology & Defect Sensitivity

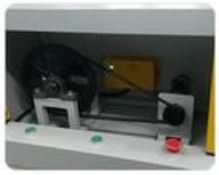
Hands-on stations for tactile defect detection, precision measurement calibration, and rapid go/no-go gauging.

KEY EQUIPMENT & RIGS:

- Vernier Caliper & Micrometer Rigs
- Blind Part Thread Sorting Rigs
- Two-Hand Symmetrical Reach Benches
- Lux-Calibrated Defect Catalog

IMPACT: **Saves 4 to 8 seconds per assembly cycle**

Zone 03 Safety Simulation & Danger Dome



Zero-Risk Experiential Hazard Pods

Sensor-interlocked danger domes where operators experience real machine hazards (pinches, shocks, strains) safely.

KEY EQUIPMENT & RIGS:

- LOTO Multi-Energy Isolation Panel
- Rotating Roller Pinch-Point Simulator
- Kinetic Spine Strain Lifting Dummy
- Chemical & UV PPE Impermeability Rig

IMPACT: **60%–80% drop in plant near-misses**

Zone 04 Standardized Work Stations & Ass



SOP Repetition & Takt-Time Pacing

Mimics actual line workstations for mastering high-torque fastening, terminal clicks, and takt synchronization.

KEY EQUIPMENT & RIGS:

- Digital Torque-Angle Nutrunner Benches
- Standard Work Combination Stations
- Wire Harness Terminal Click Rig
- Gravity Flow Bins with Poka-Yoke

IMPACT: **Guarantees 100% First-Time-Right (FTR)**

Zone 05 Kaizen & Autonomous Maintenance



Autonomous Equipment Reliability

Hands-on pneumatic circuits, sensor alignment, and daily Autonomous Maintenance (CLITA) routines.

KEY EQUIPMENT & RIGS:

- Pneumatic Cylinder Circuit Rigs
- Photoelectric & Inductive Sensor Board
- Kaizen Prototyping Workbenches
- Cutaway Pump & Valve Manifolds

IMPACT: **Reduces micro-stoppages by up to 40%**

Zone 06 Problem Solving & Digital Assess



Root Cause Analysis & Certification

Equipped with A3 boards, 5-Why root cause tools, biometric trainee logging, and digital L1–L4 skill matrices.

KEY EQUIPMENT & RIGS:

- 5-Why & Fishbone RCA Workstations
- Biometric Trainee Check-In Kiosks
- Digital Pass/Fail Stopwatch Timers
- Real-Time Competency Matrix Board

IMPACT: **100% audit pass for IATF & VDA 6.3**

MODULAR TURNKEY DELIVERY: Every zone is engineered with CE sensor interlocks, ESD protection, and plug-and-play pneumatic/electrical drops.

Layout plans custom-configured to fit plant columns, aisles, and natural lighting in 2D/3D CAD within 5 business days.

L1–L4 Capability Ladder & Simulation Catalog

OPERATOR CAPABILITY PROGRESSION MATRIX (L1 TO L4)

Structured, auditable skill certification aligned with IATF 16949 and VDA 6.3 standards:

L1

TRAINEE / NOVICE

Basic Safety & Dexterity

Demonstrates safe habits, proper PPE, blind component identification, and basic hand tool operation in simulated zone.

Off-Line Dojo Only

L2

QUALIFIED OPERATOR

Standard Work Adherence

Follows Standard Operating Procedures (SOPs) at 100% takt time pace with zero defects. Certified for independent line work.

Live Line Qualified

L3

MULTI-SKILLED TECH

Line Balancing & Flexibility

Mastered 3+ adjacent line operations, handles changeovers, performs Autonomous Maintenance (CLITA), and identifies Kaizens.

Line Relief & Flex Pool

L4

DOJO MASTER / TRAINER

Kaizen Leader & Instructor

Coaches new recruits, conducts 5-Why root-cause audits, certifies L1–L3 trainees, and leads shopfloor Kaizen initiatives.

Plant Certified Master

INDUSTRIAL SIMULATION RIGS & STATIONS WE ENGINEER

Custom fabricated at Tetrahedron's manufacturing facilities with certified industrial components:



ZONE 03: SAFETY SIMULATION

Hand Safety & Pinch-Point Danger Rig

CE sensor-interlocked rotating rollers, spring-loaded safety stops, and audible alarm simulation. Allows operators to experience simulated pinch dangers with zero injury risk.

HARDWARE: Dual optical light curtains, E-stop circuit, pneumatic dum...



ZONE 05: TPM & EQUIPMENT RELIABILITY

TPM Autonomous Maintenance Pillar Pod

Interactive learning station covering 8 TPM pillars, autonomous cleaning, inspection, lubrication, tightening, and adjustment (CLITA) protocols.

HARDWARE: Transparent cutaway gearboxes, lubrication sight-glasses, ...



ZONE 04: STANDARDIZED WORK

Digital Torque-Angle Fastening Workbench

Precision digital torque transducers measuring operator hand torque, angle feedback, cross-threading detection, and ergonomic arm fatigue positioning.

HARDWARE: Digital load-cell torque readout, Poka-Yoke screw presenta...



ZONE 02: QUALITY ASSURANCE

5-Senses Metal Sorting & Tactile Calibration

Blind sensory discrimination stations for training fingertips to identify subtle thread pitches (M6, M8, M10), burrs, surface roughness, and weight variations.

HARDWARE: Shrouded blind reach ports, precision go/no-go plug gauges...



ZONE 03: SAFETY & ERGONOMICS

PPE Matrix & Body Ergonomics Station

Comprehensive PPE selection matrix paired with a biomechanical lumbar spine strain simulator demonstrating correct lifting posture vs spine damage.

HARDWARE: Lumbar stress sensor gauge, UV eye protection booth, anti-...



ZONE 06: PROBLEM SOLVING

Problem-Solving & A3 RCA Workstation

Structured root cause analysis whiteboard with magnetic Ishikawa fishbone, 5-Why ladders, and digital stopwatch test logging for L1–L4 certification.

HARDWARE: Magnetic RCA boards, digital cycle timer, trainee biometri...

CUSTOM SIMULATION BENCHES: We engineer tailored simulation rigs for automotive wiring, EV battery module assembly, precision CNC machining, sheet metal press shops, plastic injection molding, and pharma packaging.

8-Week Turnkey Implementation Roadmap

END-TO-END PROJECT GOVERNANCE FROM DEFECT AUDIT TO FULL LAUNCH

Tetrahedron delivers a complete turnkey service. We do not just provide drawings or equipment; our master consultants analyze your plant scrap pareto, manufacture the simulation benches, author bilingual training standard operating procedures, and certify your internal trainers.

01 STEP — WEEKS 1 – 2

Needs Assessment & Scrap Pareto Audit

Comprehensive on-site plant diagnostic: analyzing assembly bottlenecks, customer rejection logs, operator turnover rates, and specific line skill gaps.

MILESTONE DELIVERABLE:
Output: Scrap Pareto & Critical Process Map

02 STEP — WEEK 3

Conceptual Design & 3D CAD Floor Architecture

Architectural planning of the Dojo room: station footprints, ergonomic material flow, electrical/pneumatic drop routing, visual management zones.

MILESTONE DELIVERABLE:
Output: Approved 3D CAD Floor Plan & Rig Specs

03 STEP — WEEKS 4 – 5

Physical Rig Fabrication & CE Interlocks

Heavy-duty steel & aluminum profile fabrication of simulation benches, sensorized pinch-point danger domes, and digital torque stations in our factory.

MILESTONE DELIVERABLE:
Output: Commissioned Simulation Jigs & Benches

04 STEP — WEEK 6

Content Development & Bilingual SOS/SOPs

Authoring standardized training work instructions (SOS), pass/fail visual criteria, evaluation sheets, and digitized L1–L4 competency rubrics.

MILESTONE DELIVERABLE:
Output: Visual SOP Manuals & Grading Rubrics

05 STEP — WEEK 7

Train-The-Trainer (TTT) Master Accreditation

Intensive pedagogical coaching and technical certification of internal plant trainers, team leaders, and shift supervisors to run experiential drills.

MILESTONE DELIVERABLE:
Output: Certified Internal Dojo Masters

06 STEP — WEEK 8

Pilot Batch Commissioning, Rollout & Handover

Facilitating the maiden cohort of plant trainees through the Dojo, tuning cycle times, establishing HR/operations integration, and tracking initial KPIs.

MILESTONE DELIVERABLE:
Output: Live Plant Capability Academy Handover

FINANCIAL BUSINESS CASE: CAPITAL PAYBACK IN 3 TO 6 MONTHS

Representative cost-benefit model calculated across 100+ commissioned manufacturing plants:

SCRAP REDUCTION

70% – 90%

Elimination of training defects on live production parts. Operators fail on reusable simulator blocks instead of high-value client inventory.

ONBOARDING SPEED

80% FASTER

Operator cycle time readiness reduced from 30–45 days of erratic shadow training down to 5–7 days of focused tactile muscle memory.

DEFECT CONTAINMENT

ZERO DEFECTS

Zero defect leakage to OEM customers. First-Time-Right (FTR) assembly rates reach 99.8%+ within the first operational quarter.

SAFETY & ACCIDENTS

60%–80% DROP

Near-miss incident drop through sensory calibration in the Danger Dome. Lower medical insurance liabilities and zero lost-time injuries.

EXECUTIVE ROI PAYBACK SUMMARY

Average Plant Investment	Turnkey Dojo setup costs range between INR 15 Lakhs to 45 Lakhs depending on square footage (600–2000 sq.ft) and hardware simulation density.
Monthly Scrap & Defect Savings	A plant producing 10,000 units/day with a 1.5% novice defect rate saves INR 3.5 to 8 Lakhs per month immediately upon Dojo commissioning.
Net Payback Horizon	Full capital expense recovery is achieved between Month 3 and Month 6 post-rollout, generating compounding productivity gains for 5+ years.
Tier-1 OEM Audit Premium	Guarantees full qualification marks during Maruti Suzuki, Tata Motors, Hero MotoCorp, VDA 6.3, and IATF 16949 supplier quality audits.

Client Track Record & Turnkey Consultation

100+
Dojos Commissioned
Across India & Global Plants

50,000+
Workers Certified
L1–L4 Competency Accredited

100%
Audit Pass Rate
IATF 16949 & VDA 6.3 Ready

TRUSTED BY 30+ ENTERPRISE MANUFACTURING LEADERS

Powering workforce capability academies for automotive OEMs, Tier-1 giants, and heavy engineering plants:

ADVIK · PADMINI VNA MECHATRONICS · Lucas-TVS Ltd · HARSHA · DELUX · Frencken · TRITON · CMR For a better tomorrow · HÖRMANN · TATA MOTORS Connecting Aspirations · CLYDEUNION PUMPS · BLUE STAR · Coromandel · MINDA Minda SAI Ltd. · AASRAA · MSP GROUP · ADM · AIS Asahi India Glass Ltd.

Hero MotoCorp · Royal Enfield · Tata Motors · Autoliv · Mahindra CIE · Blue Star · Minda Group · SKH · Schneider Electric · Subros

THE COMPLETE TETRAHEDRON TURNKEY DELIVERABLE PACKAGE

Everything required for Day-1 operational handover is included in our project scope:

- [OK] **Spatial 3D CAD Floor Plans**
  Customized layout drawings, 3D renderings, and utility drop schematics.
- [OK] **Industrial Simulation Benches**
  CE-compliant tactile simulation rigs with sensor interlocks and safety shields.
- [OK] **Bilingual SOP & SOS Manuals**
  Standardized work instruction sheets with photographic pass/fail rubrics.
- [OK] **L1–L4 Competency Software**
  Operator skill tracking templates, scorecards, and biometric logging.
- [OK] **Train-The-Trainer Certification**
  Comprehensive pedagogical coaching for 4–8 internal plant supervisors.
- [OK] **1-Year On-Site Support Warranty**
  Annual maintenance, jig recalibration, and refresher trainer audits.

START YOUR PLANT DOJO TRANSFORMATION

Book a 1-on-1 Virtual Consultation with Our Principal Dojo Architects

Get a tailored scrap Pareto evaluation, conceptual 3D layout recommendations, and a formal commercial proposal calibrated for your plant's assembly requirements.

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